

IPL MATCHES DATA ANALYSIS REPORT

1. INTRODUCTION

The Indian Premier League (IPL) is one of the most popular and competitive T20 cricket leagues in the world. Since its launch in 2008 by the Board of Control for Cricket in India.

The tournament features franchise-based teams representing different cities of India and includes some of the best players from around the world.

2. OBJECTIVE

The main objective of this IPL Data Analysis project is to analyze historical IPL match data to discover meaningful insights related to team performance, player achievements, match outcomes, toss decisions, venue advantages, and seasonal trends.

- Understanding team strategies
- Evaluating player performance
- Improving match predictions
- Supporting decision-making for teams and analysts

3. PROBLEM STATEMENT

- Which IPL has been the most successful across all Seasons?
- Which trends and patterns can be identified from historical IPL match data?
- Which teams perform better while batting first or chasing targets?
- Which stadiums host the highest no. of IPL matches?

4. DATASET DESCRIPTION

The IPL dataset includes the following important features:

- **Season** – IPL season or year in which the match was played
- **Date** – Date on which the match took place
- **City** – City where the match was conducted
- **Venue** – Stadium or ground where the match was played
- **Team1** – First competing team in the match
- **Team2** – Second competing team in the match
- **Toss Winner** – Team that won the toss
- **Toss Decision** – Decision taken after winning the toss (Bat or Field)
- **Winner** – Team that won the match
- **Result** – Final result status of the match
- **Win by Runs** – Margin of victory measured in runs
- **Win by Wickets** – Margin of victory measured in wickets
- **Player of Match** – Best performing player of the match
- **Target Runs** – Runs required by the chasing team to win
- **Target Overs** – Overs available to chase the target
- **Method** – Match result method (for example, D/L method if applied)

5. DATA CLEANING AND PREPROECSSING

Before visualization, the dataset was cleaned to ensure accuracy:

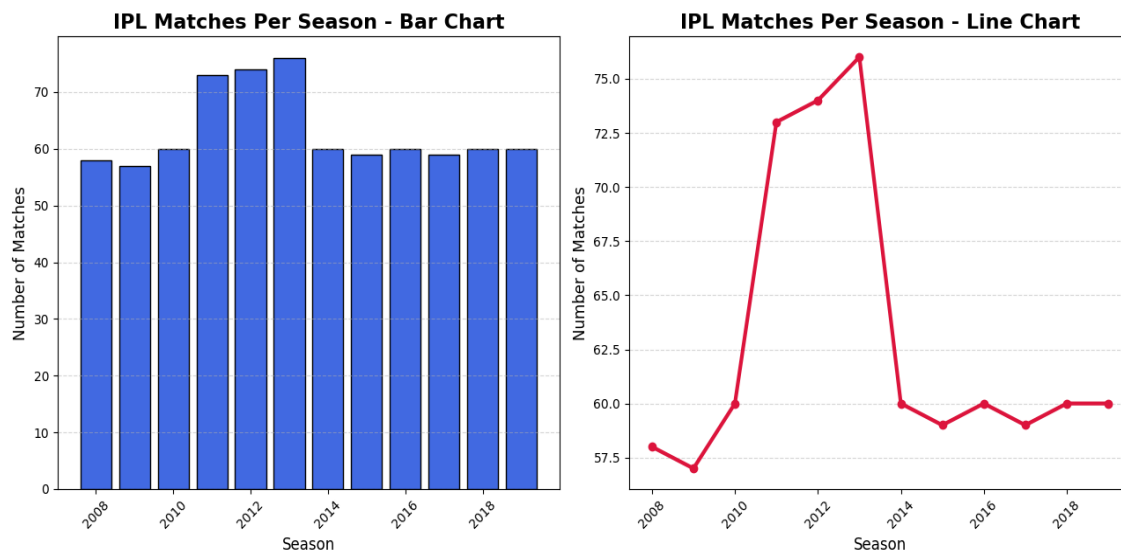
- Handled missing values:
- Numerical → replaced with mean
- Categorical → replaced with mode
- Converted data types where necessary
- Ensured no null values remained

6. TECHNOLOGY USED IN:-

- Python libraries
 - Pandas
 - Matplotlib
 - Seaborn
- Jupiter notebook
- Google sheets

7. DATA VISUALIZATION AND ANALYSIS

1. IPL MATCHES PER SEASON



The above charts represent the total number of IPL matches played in each season from 2008 to 2019. Both the bar chart and line chart help visualize the yearly trend and growth of the tournament over time.

The analysis shows that between 2008 and 2010, around 57–60 matches were played each season. A significant increase was observed in 2013, when approximately 76 matches were played during the season. However, after 2013, the number of matches dropped to around 60 per season and remained mostly consistent in the later years.

This indicates that the IPL management standardized the tournament structure after experimentation with expanded formats in earlier seasons.

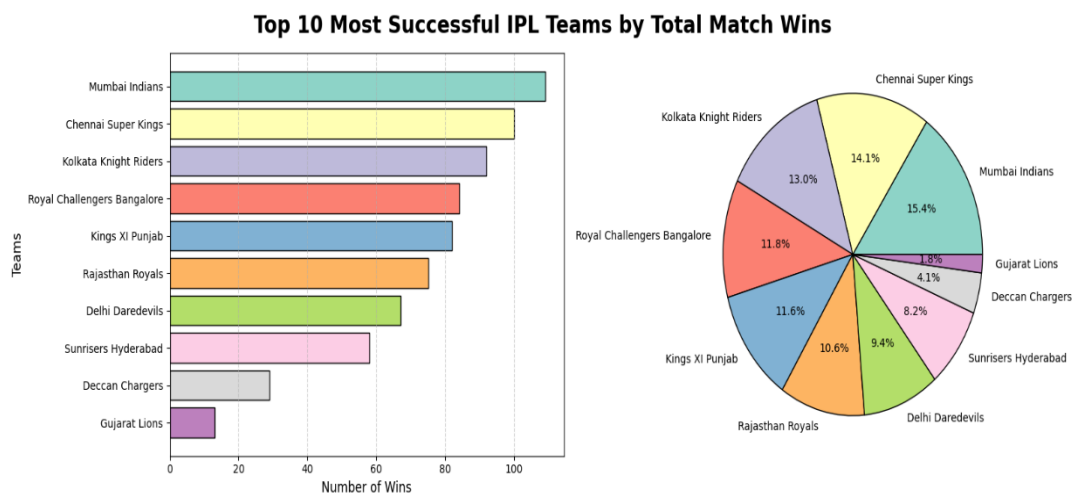
Why we use this chart: -

1. Because The **bar chart** is useful for comparing the number of matches between seasons, making it easy to identify seasons with the highest or lowest number of matches.
2. And The **line chart** helps visualize the overall trend and fluctuations in match count over time, showing periods of growth, expansion, or stability in the tournament.

Insights: -

1. IPL experienced major growth between 2011 and 2013.
2. The highest number of matches was played in 2013.
3. Later seasons maintained a more stable tournament format.
4. The charts clearly show the evolution and consistency of the IPL over time.

2. TOP 10 IPL TEAM BY WINS



The Second charts represent that 10 most successful teams based on their total number of match wins

The analysis shows that Mumbai Indians and Chennai super kings have the most successful IPL franchise with the highest number of total wins. on the other hand, teams like Deccan charges and Gujarat lions recorded the lowest no. of wins because they participated in fewer IPL seasons as compared to other franchises

The visualization also helps in understanding the relative performance difference between franchises.

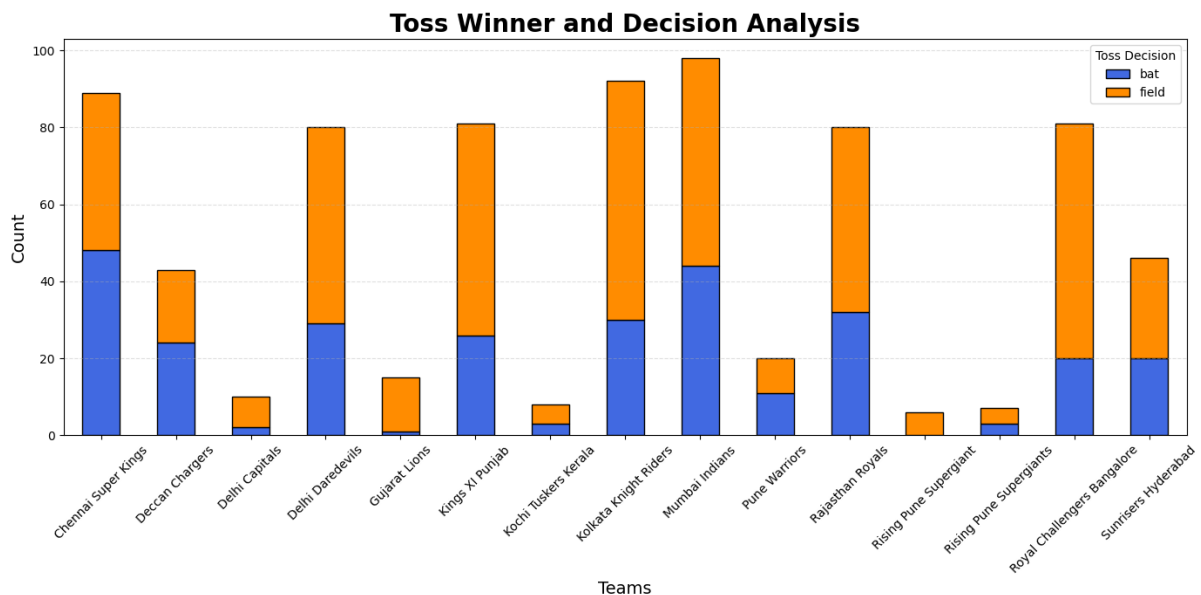
Why we use this chart: -

1. The bar chart provides a clear comparison of the number of wins achieved by each team.
2. the pie chart shows the percentage contribution of each team to the total wins among the top 10 teams.

Insights: -

1. Mumbai Indians emerged as the most successful IPL team.
2. Chennai Super Kings maintained consistently strong performances.
3. Kolkata Knight Riders secured a strong position among top teams.
4. Some teams had lower win counts due to fewer tournament appearances.
5. The charts effectively show both comparison and percentage contribution of team victories.

3. TOSS WINNER AND DECISION ANALYSIS



The Third chart represents the Toss decision made by different IPL teams after winning

The analysis shows that most IPL teams preferred choosing to **field first** rather than bat first after winning the toss. Teams such as **Mumbai Indians**, **Kolkata Knight Riders**, **Royal Challengers Bangalore**, and **Chennai Super Kings** selected the fielding option more frequently

This trend indicates that teams generally preferred chasing targets, possibly due to pitch conditions, dew factors, and the advantage of knowing the target score in T20 cricket.

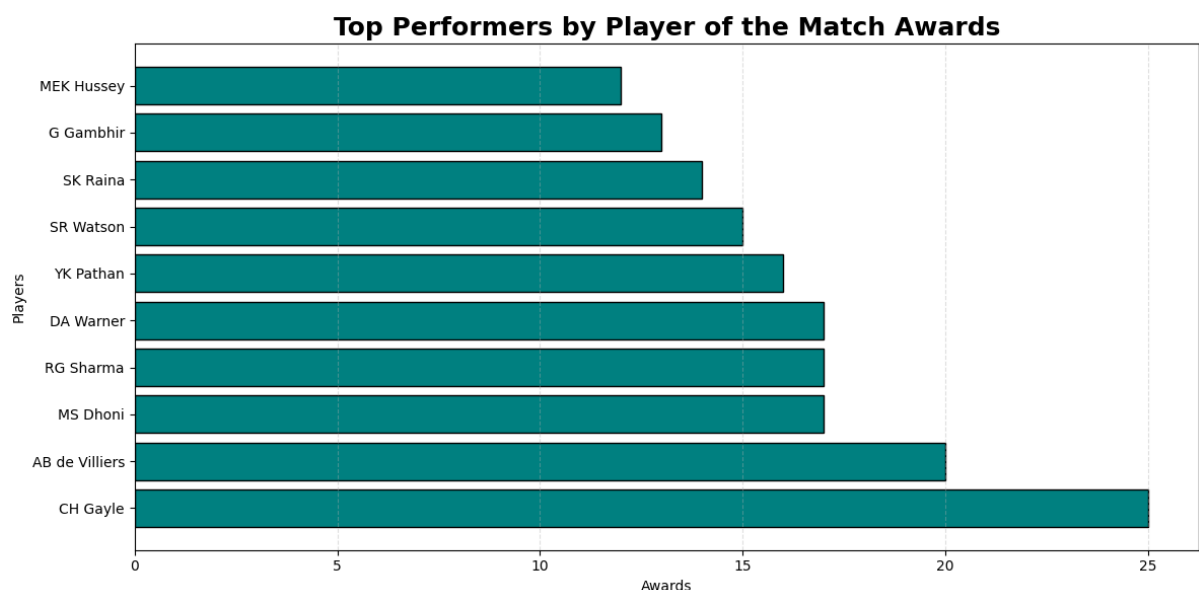
Why we use this chart: -

1. Because the stacked bar chart compares how many times each team chose to **bat first** or **field first** after winning the toss. Also the chart clearly shows the strategic difference between batting-first and chasing approaches.

Insight: -

1. Most IPL teams preferred fielding first after winning the toss.
2. Chasing targets appears to be a common strategy in IPL matches.
3. Mumbai Indians and Kolkata Knight Riders had high toss-winning frequencies.
4. Teams with fewer IPL appearances had lower toss decision counts.

4. TOP PERFORMERS BY PLAYER OF THE MATCH AWARDS



The Fourth chart focuses on the players who received the highest no. of players of the match awards in IPL history and also helps to identify the most influential and match winning players across different IPL season

The analysis generally shows that legendary IPL players such as **Chris Gayle**, **AB de Villiers**, **MS Dhoni**, **Rohit Sharma**, and **Virat Kohli** frequently appeared among the top performers due to their consistent impact over multiple

seasons. Their ability to deliver match-winning performances helped their teams achieve success in the tournament.

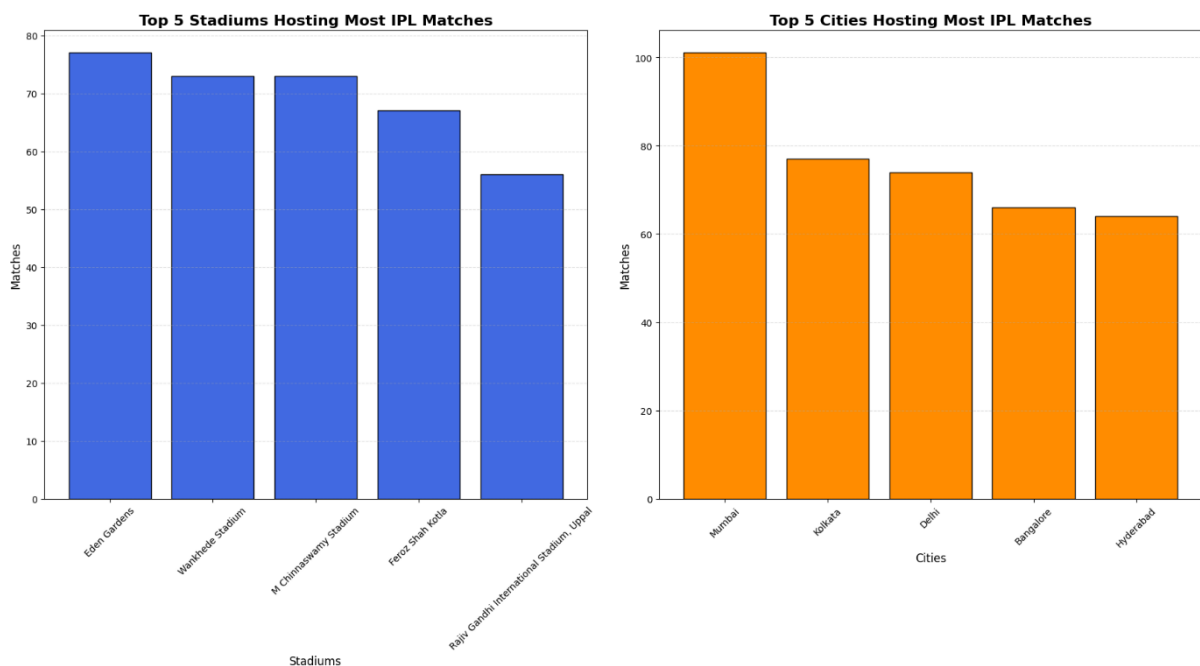
Why we use this chart:-

1. Because the bar chart provides of the no. of player of match achieved by players

Insight :-

1. Top players consistently received multiple Player of the Match awards.
2. Match-winning players played important roles in team success.
3. Consistent performers maintained high impact across seasons.

5. TOP 5 STADIUMS AND CITIES HOSTING MOST IPL MATCHES



The fifth chart represent the top 5 stadiums and cities that hosted the highest number of IPL matches. The bar charts help compare the number of matches played at different stadiums and across major IPL host cities.

Also, the charts clearly show how certain cities and stadiums became major centres for IPL cricket over the years. These venues not only hosted regular league matches but also important playoff and final matches in several seasons.

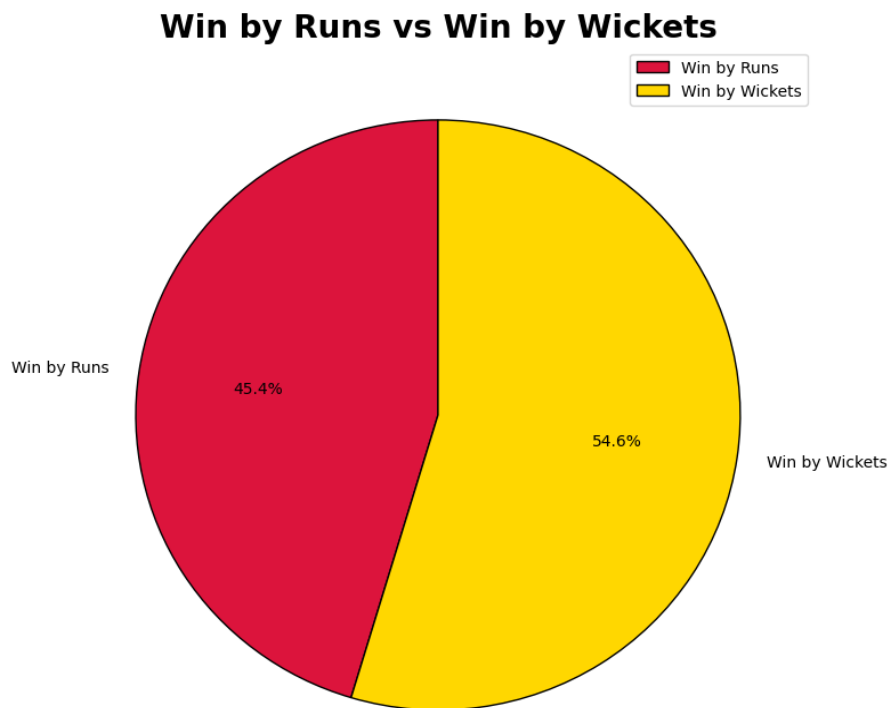
Why we use this chart: -

1. The bar charts help compare the number of matches played at different stadiums or cities. And clearly show differences between categories

Insights: -

1. Eden Gardens hosted the highest number of IPL matches among stadiums.
2. Mumbai was the city with the highest number of hosted IPL matches.
3. Wankhede Stadium and M. Chinn swamy Stadium were also major IPL venues.
4. Major metropolitan cities dominated IPL hosting due to better infrastructure and fan support.
5. The charts highlight the importance of iconic cricket stadiums in IPL history.

6. WIN BY RUNS AND WIN BY WICKETS



The above pie chart represents the comparison between IPL matches won by **runs** and matches won by **wickets**. It helps in understanding the most common type of match victory in IPL history.

The analysis shows that approximately **54.6%** of IPL matches were won by **wickets**, while around **45.4%** of matches were won by **runs**. This indicates that chasing teams have been slightly more successful than teams batting first in IPL matches.

The higher percentage of wins by wickets suggests that teams often prefer chasing targets in T20 cricket. Factors such as pitch conditions, dew effect, and knowing the target score may provide an advantage to the chasing team.

On the other hand, matches won by runs indicate successful defensive performances where teams effectively protected their target scores.

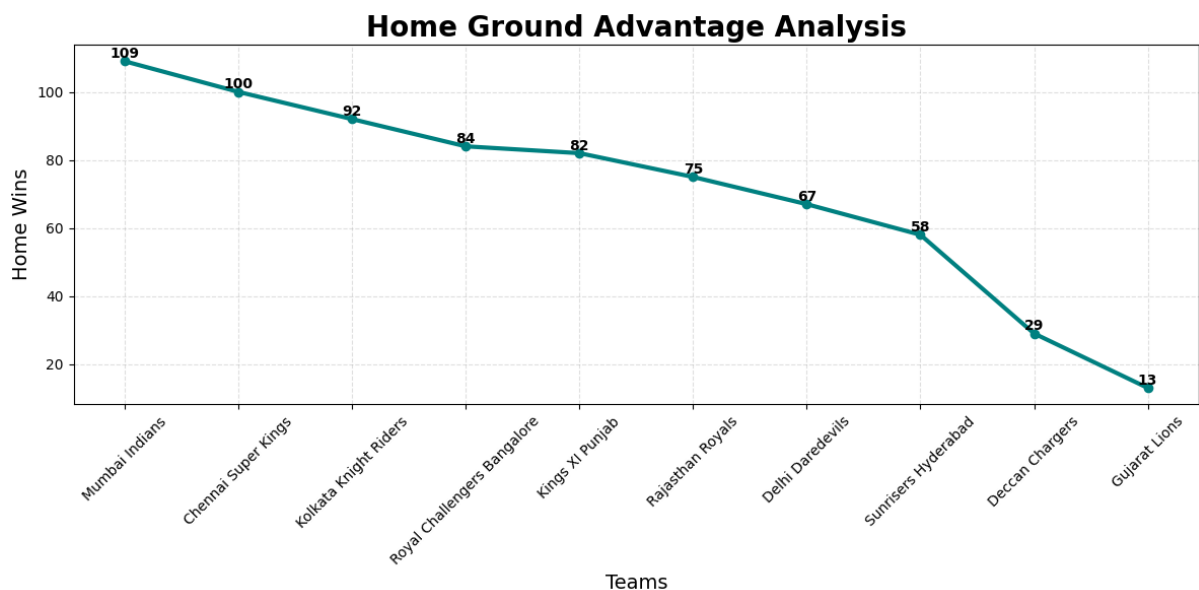
Why we use this chart: -

1. The pie chart clearly highlights the difference between both types of victories and helps understand match outcome trends in IPL tournaments. And also, to understand whether batting first or chasing is more successful

Insight: -

1. Wins by wickets are slightly higher than wins by runs.
2. Chasing teams had a better success rate in IPL matches.
3. Around 54.6% of matches were won while chasing.
4. Approximately 45.4% of matches were defended successfully.

7. HOME GROUND ADVANTAGES ANALYSIS



The seventh chart represents the number of matches won of IPL teams on their home ground. Also that chart clearly demonstrates that most successful IPL teams performed strongly at their home grounds, highlighting the importance of venue familiarity and home support in T20 cricket.

The analysis shows that **Mumbai Indians** recorded the highest number of home wins with approximately 109 victories, followed by **Chennai Super Kings** with around 100 home wins. This indicates that both teams have been extremely strong and consistent while playing at their home venues.

Teams such as **Kolkata Knight Riders**, **Royal Challengers Bangalore**, and **Kings XI Punjab** also achieved a significant number of home victories, showing the importance of familiar pitch conditions, local crowd support, and stadium experience in IPL matches.

Why we use this chart:-

1. The **line chart** helps visualize the overall trend and fluctuations in match count over time, showing periods of growth, expansion, or stability in the tournament.